MIL-P-14574B(MR)
5 June 1984
SUPERSEDING
MIL-P-14574A(MR)
1 May 1964

MILITARY SPECIFICATION

PACKING (WIPER RING), SYNTHETIC RUBBER, CLOTH INSERTION

This specification is approved for use by Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of synthetic rubber packing, conforming to M1BC 610 of ASTM D2000, used on recoil mechanisms as a wiping ring for the recoiling gun type.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and Standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Shipping, Fiberboard
PPP-F-320 - Fiberboard, Corrugated And Solid, Sheet Stock (Container Grade) And Cut Shapes
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY
MIL-B-117 - Bag, Sleeve And Tubing - Interior Packaging
MIL-P-14232 - Parts, Equipment And Tools For Army Material, Packaging Of

STANDARDS

FEDERAL
FED-STD-191 - Textile Test Methods

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

American Society for Testing and Materials (ASTM) Standards

ASTM D 2000 - Classification System for Rubber Products in Automotive Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They also are distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) and before production has commenced, samples of the rubber compound and of the fabric to be used in fabricating and molding the packing shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. A certification to the effect that the final packing shall comply with the requirements of this specification shall accompany the samples. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification.

3.2 Materials.

3.2.1 Rubber compound. The rubber compound shall conform to M1BC 610 of ASTM D 2000).

3.2.2 Fabric. The fabric shall be cotton duck, evenly and firmly woven; free from unsightly defects, dirt, knots, holes, weak places, bumps, and irregularities of twist, and consistent in all respects with good manufacturing practice. The fabric shall be suitable for frictioning (impregnation) with synthetic rubber and when tested as specified in 4.8.2 shall conform to the physical properties specified in table I.

TABLE I. Cotton duck

Weight per square yard (minimum)	Thread Count		Gauge (inches) (min.)	Breaking strength (4 inch width) pounds (force)	
	Warp (min.)	Filling (min.)		Warp (min.)	Filling (min.)
10 ounces	32	28	.031	135	100

3.3 Construction. The packing shall consist of layers of cotton duck fabric impregnated with a rubber compound. Six to 10 plies of fabric shall be uniformly distributed throughout the rubber compound. This construction shall be press cured to the shape specifed in Figure 1 (see 6.2 for applicable drawing) so that the compound and fabric become an integral part of each other.

3.4 Coating. The packing shall be given a smooth surface treatment with a high melting point wax or mixture of waxes, and a coating of powdered graphite. There shall be no surface contamination.

3.5 Workmanship. The packing shall be free of roughness, mold marks, pits, tears, cracks or breaks, and the presence of foreign particles or grit (see 4.4).

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Note: All dimensions in inches.

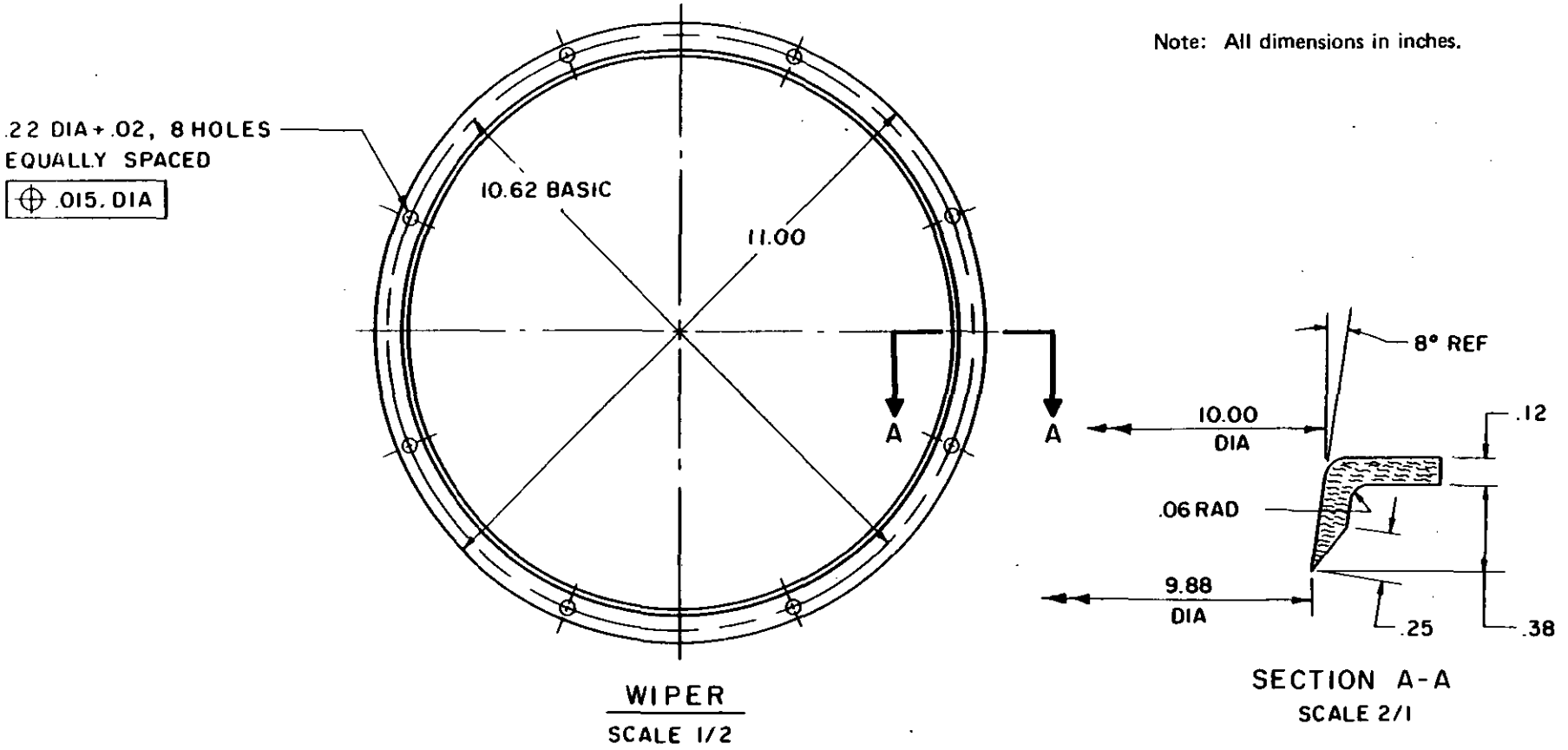


FIGURE 1. Packing shape.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall be performed on samples when first article samples are required (see 3.1 and 6.2). This inspection shall include the examination of 4.7, in which samples shall be free from the defects listed in tables III and IV, and the tests of 4.8. The first article may be a standard production item from the contractor's inventory provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing techniques applicable to the remaining units to be furnished under the contract. In addition to the first article, samples shall be submitted for performing all tests required by this specification.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on the samples selected in accordance with 4.6. This inspection shall be as specified in table II.

TABLE II. Quality conformance inspection

Examination or test	Requirement paragraph	Examination or test - paragraph and table
Rubber compound Physical properties	3.2.1	4.8.1
Fabric Visual, weight dimensional, breaking point	3.2.2	4.7.1, 4.8.2, table I and table IV
Packing Visual and dimensional	3.3, 3.4 and 3.5	4.7 and table III

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4.5 Lot. Unless otherwise specified, a lot shall consist of all packings produced under essentially like conditions from the same lot of raw materials and offered for acceptance at one time.

4.6 Sampling

4.6.1 Sampling for inspection. Sampling for inspection of packing for defects listed in table III shall be in accordance with MIL-STD-105.

4.6.2 Sampling for tests: Samples of rubber and fabric that are representative of the lot shall be selected for tests. Quantities shall be as follows:

(a) Rubber: Samples of rubber compound shall consist of 3 pads, 6 by 6 $\pm$ 0.2 inch by 0.075 $\pm$ 0.002 inch, and 4 plugs 1.129 $\pm$ 0.005 inches in diameter and 0.500 $\pm$ .010 inch thick.

(b) Fabric: A sample of the duck shall consist of 1 linear yard.

4.7 Examination. Sample units of packing selected in accordance with 4.6.1 shall be examined for the defects and at the acceptable quality levels shown in table III.

4.7.1 Examination of fabric. A sample of the fabric shall be examined for suitability for frictioning in accordance with the requirements of 3.2.2.

4.7.2 Examination of packaging, packing and marking. Examination shall be made to ascertain that the packaging, packing and marking of the material conform to this specification.

4.8 Tests. All tests of this specification shall be required on both the rubber compound (see 4.8.1) and the fabric (see 4.8.2) for both the first article sample inspection and acceptance inspection.

4.8.1 Tests of rubber compound. Samples of rubber compound selected in accordance with 4.6.2 shall be tested in accordance with the applicable requirements and tests of Standard ASTM D2000.

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TABLE III. Classification of defects

Category	AQL percent	Defect	Method of Inspection
Critical		None defined	
Major 101	1.0	Wrong number of fabric plies (see 3.3)	Visual
Major 102		Plies not uniformly distributed (see 3.3)	Visual
Major 103		Wrong shape (see 3.3) or dimensions (see 6.2)	Measure
Major 104		Compound and fabric not intregal part of each other (see 3.3)	Visual
Major 105	2.5	Improper coating (see 3.4)	Visual
Major 106		Incomplete coating (see 3.4)	Visual
Major 107		Surface contaimination (see 3.4)	Visual
Major 108	1.0	Roughness (see 3.5)	Visual
Major 109		Mold marks (see 3.5)	Visual
Major 110		Pits (see 3.5)	Visual
Major 111		Tears (see 3.5)	Visual
Major 112		Cracks or breaks (see 3.5)	Visual
Major 113		Indicated presence of foreign particles of grit (see 3.5)	Visual

4.8.2 Tests of fabric. Cotton duck selected as specified in 4.6.2 shall be tested in accordance with table IV. Unless otherwise specified the method of test in table IV shall be that of Fed. Test Method Std. No. 191.

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TABLE IV. Methods of test for fabric

Test	Requirement	Test Method	Number of Determinations	Required as
Weight	Table I	5041	5	Avg of the 5 determinations to nearest 0.01 ounce
Thread count:				
Warp	Table I	5050	5	Avg of the 5 determinations to nearest whole number
Filling	Table I	5050	5	Avg of the 5 determinations to the nearest whole number
Gauge	Table I	Micrometer	5	Avg of the 5 determinations to the nearest .001 inch
Breaking strength:				
Warp	Table I	5100	5	Avg of the 5 determinations to the nearest 1.0 pound
Filling	Table I	5100	5	Avg of the 5 determinations to nearest 1.0 pound

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Two wiper blades separated by an 11.5 inch square fiberboard stiffener conforming to PPP-F-320, Type CF, Grade 125, shall be unit packed in a 13 inch square bag conforming to MIL-B-117, Type 1, Class C. Closure shall be accomplished by heat seal, staples or tape. Intermediate containers, each containing ten unit packs, shall be of fiberboard in accordance with PPP-B-636, Style RSC, Grade W5C of interior dimensions 13.5 X 13 X 11.5 inches. Closure shall be accomplished with two pieces of tape (2 X 16 inches) conforming to PPP-T-76.

5.1.2 Level C. Wiper blades shall be preserved in accordance with MIL-P-14232, two per unit pack.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2) and in accordance with MIL-P-14232.

5.3 Marking. Unit, intermediate and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers one type of synthetic rubber packing used on recoil mechanisms as a wiping ring for the recoiling gun type.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Title, number and date of applicable drawing (see 3.3).
(c) Number of packings (wiper rings) required.
(d) First Article: Whether first article and certified report are required (see 3.1).
(e) Level of preservation, packaging and packing required (see 5.1 and 5.2).

6.3 Figure 1 is to designate shape only (see 6.2(b)).

6.4 Metric units. When metric units are required, units for inch, yards, ounces and pounds may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	2.54	=	centimeter (cm)
yard	0.9144	=	meter (m)
ounce	28.350	=	gram (gr)
pound(force)	4.448	=	newton (N)

Note: Conversion factors can be associated with ASTM E 380 entitled "Metric Practice Guide."

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Review activities:

Army - AR
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Preparing activity:

Army - MR

Project No. 5330-A093

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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